

Emergent Attention: Iterated Local Neural Cellular Automata as a Noise-Robust Token Mixer for Vision Transformers

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Abstract

We introduce **Emergent Attention**, a drop-in replacement for multi-head self-attention (MHSA) that replaces $O(n^2)$ all-to-all token mixing with K iterations of a local Neural Cellular Automaton (NCA) operating on the patch grid. Each iteration applies a **fixed** (non-learned) 3×3 depthwise perception (Sobel-X, Sobel-Y, Identity) followed by a shared two-layer MLP producing an additive, Bernoulli-gated cell update. We propose **Hybrid NCA-ViT**: the first half of the backbone uses Emergent Attention, the second half uses standard MHSA. On CIFAR-100, it reaches **81.91% top-1** with 6.46M parameters — matching Swin-Tiny (81.56%, 27.60M params, $4.3\times$ more parameters) and beating DeiT-Tiny by **+9.0 points** under identical training.

The paper's central contribution is not this clean-data number — Swin matches it with far more parameters — but a **specific, mechanistically-grounded robustness property**. On CIFAR-100-C (19 corruption types $\times$ 3 severities), Hybrid NCA-ViT has the highest mean corrupted accuracy of five architectures, and the advantage is **not uniform**: it concentrates almost entirely in additive high-frequency pixel noise (+8.3pp over the best attention-only baseline) and is negligible-to-negative on blur, weather, and impulse noise. We verify this is not an artifact by measuring representation drift directly: the classifier's pre-head feature moves *less* under corruptions where the hybrid wins and *more* under the one family where it loses — a sign-correct falsification test. We validate the prediction on a real, non-synthetic domain: **BUSI, 780 breast-ultrasound images**, where speckle — the dataset's dominant noise process — is exactly the corruption family the mechanism favors. Across 3 seeds, the hybrid beats a matched DeiT-Tiny on balanced accuracy (49.8% vs. 36.4%) and macro-F1 (41.5% vs. 21.0%) with **no overlap in range between architectures**.

Keywords: vision transformer, neural cellular automata, corruption robustness, representation stability, medical imaging, data efficiency.

1 Introduction

Self-attention is the computational core of the modern Vision Transformer (ViT) [1]. Every token attends to every other token in a single layer, so global context is available immediately, at $O(n^2)$ cost in the number of tokens n . A complementary line of work, **Neural Cellular Automata** (NCA) [4], shows that complex spatial behavior can emerge from repeated application of a small, local update rule — a computational regime that is intrinsically local, translation-equivariant, and iteratively refining, properties softmax attention lacks by construction.

Two recent papers combine NCA and Transformer machinery. **AdaNCA** [12] (NeurIPS 2024) inserts NCA blocks as adaptors *between* existing attention layers to improve robustness; standard MHSA remains the primary mixer throughout. **ViTCA** [13] (NeurIPS 2022) fuses attention *into* the NCA cell-update rule itself. In

both, attention is never removed. We ask a more direct question:

Can iterated local NCA dynamics fully replace global softmax attention as the token mixer, rather than augment it — and if the resulting model is more robust to noise, can that robustness be characterized precisely enough to predict where it will and will not appear?

Our answer to the first half is yes, with a specific architecture: NCA blocks for early local feature formation, standard attention blocks for late global aggregation ("Hybrid NCA-ViT"). Our answer to the second half is the paper's central contribution: the robustness is **not general** — it concentrates in additive high-frequency noise, is measurable directly in the network's internal representations, and correctly predicts its own failure mode (impulse noise). We validate the prediction on a real domain, ultrasound imaging, chosen specifically because its dominant artifact (speckle) is the corruption family the mechanism favors.

Contributions

- (1) **Emergent Attention** — an NCA-based token mixer with fixed, non-learned perception, a zero-initialized update MLP, and Bernoulli-gated updates. Unlike AdaNCA/ViTCA, this **fully replaces** MHSA rather than adapting or fusing with it.
- (2) **Hybrid NCA-ViT**, with a two-seed depth-ordering ablation (§5.2) showing NCA-early/attention-late is a genuine accuracy peak, not simply "using both mixers."
- (3) A **frequency-specific characterization of corruption robustness** (§5.4): the advantage over attention-only baselines concentrates in additive pixel noise (+8.3pp), measured across all 19 CIFAR-100-C corruption types.
- (4) A **mechanistic, falsifiable explanation** (§5.5): representation drift is lower for the hybrid on 17/19 corruptions — and higher on the one family where it loses, matching the theory's predicted failure mode.
- (5) **Real-domain validation on BUSI ultrasound** (§5.7), chosen because speckle matches the mechanism directly. Verified across 3 seeds with non-overlapping performance ranges.
- (6) A **reproducible implementation** — configs, trained checkpoints, and every evaluation/analysis script released.

2 Related Work

Vision Transformers. ViT [1] established that pure attention stacks can match convolutional networks at scale. DeiT [2] closed the data-efficiency gap with distillation and heavy augmentation. Swin [3] reintroduced locality via shifted windows. We take locality further — to a per-cell 3×3 neighborhood — and recover globality through iteration rather than windowing.

Neural Cellular Automata. NCA [4] showed that a small local update rule, applied repeatedly, reproduces target images from a single seed cell. Subsequent work extended NCAs to classification, texture synthesis, and volumetric growth.

NCA combined with Transformers — closest prior work. AdaNCA [12] inserts NCA adaptors between ViT attention layers and reports improved robustness across 8 benchmarks, including "certain types of noise" — the published description does not decompose which corruption families benefit or characterize a mechanism, and attention remains the primary mixer throughout. ViTCA [13] computes the NCA's per-cell update via self-

attention over a local neighborhood — attention is fused into the cellular update rule, not replaced. Emergent Attention differs along three axes: **(i) full replacement, not augmentation** — no attention computation occurs inside NCA blocks; **(ii) fixed, non-learned perception** by default; **(iii) a frequency-specific, mechanistically-verified robustness claim** (§5.4–5.5) rather than an aggregate one — to our knowledge neither prior work provides a representation-level causal account of why NCA helps or predicts which corruptions it will fail on.

Adversarial and corruption robustness. FGSM [10] and PGD [11] measure worst-case robustness; Hendrycks & Dietterich's CIFAR-100-C [14] measures robustness to common, naturally-occurring corruptions at graded severity — the benchmark we use for the frequency decomposition in §5.4, precisely because it separates corruption *types* rather than reporting one aggregate number.

3 Method

3.1 Preliminaries

A standard ViT block computes:

$$z' = z + \text{MHSA}(\text{LN}(z)) , \quad z = z' + \text{MLP}(\text{LN}(z'))$$

We retain this residual skeleton and replace $\text{MHSA}(\cdot)$ with $\text{NCAAttention}(\cdot)$.

3.2 Emergent Attention

Let $z \in \mathbb{R}^{B \times (l+n) \times D}$ be the token sequence, CLS at index 0, patch tokens at indices 1..n, arranged on an $H_p \times W_p = n$ grid.

CLS isolation. The CLS token never participates in the NCA computation — the block returns a zero delta for it, preventing CLS from acting as an information sink inside a rule meant to be shared identically across all spatial cells.

Sequence $\rightarrow$ grid.

$$s_0 = \text{rearrange}(\text{LN}(p), "b (h_p w_p) d \rightarrow b d h_p w_p")$$

Perception. A fixed depthwise 3×3 convolution (groups=D) applies M=3 hand-chosen filters per channel:

-1	0	1
-2	0	2
-1	0	1

Sobel-X

-1	-2	-1
0	0	0
1	2	1

Sobel-Y

0	0	0
0	1	0
0	0	0

Identity

These filters are frozen by default (`learnable_filters=False`); an optional flag unfreezes them but is not used for any headline result in this paper.

Update MLP (shared across all cells):

$$h = \text{ReLU}(W_1 \cdot \text{perc} + b_1) , \quad \delta = W_2 \cdot h + b_2$$

W_2, b_2 are **zero-initialized**, so $\delta \equiv 0$ at step 0 and the block is an exact identity map at the start of training — no NCA-specific warm-up schedule is required.

Stochastic update. At each of K iterations:

$$s_{t+1} = s_t + m_t \odot \delta_t, \quad m_t \sim \text{Bernoulli}(p) \quad (\text{training})$$

$$s_{t+1} = s_t + \delta_t \quad (\text{inference})$$

default $p = 0.5$, following standard NCA stabilization practice [4]

Output. After K iterations (default $K=4$), the block returns the accumulated delta $\Delta = s_K - s_0$, reshaped to a sequence, concatenated with a zero CLS delta, and added to the residual stream. Figure 1(b) diagrams one iteration end-to-end.

3.3 Architectures

Hybrid NCA-ViT. The first $L_N=6$ blocks use Emergent Attention ($D=192$, $K=4$, $H_{\text{mlp}}=384$); the remaining $L_A=6$ use standard MHSA (3 heads). Classification uses **global average pooling over patch tokens** — CLS is excluded from the NCA grid throughout the network and is never populated with image-dependent information, so the head reads patch tokens directly. Total: **6.46M parameters**.

NCA-ViT (pure). All 12 blocks use Emergent Attention. Total: 7.31M parameters. Used as an isolating baseline in §5.2 and §5.5 — the attribution argument in §5.5 rests on this model's 7.9-point clean-accuracy deficit against Swin, yet still winning on several corruption types.

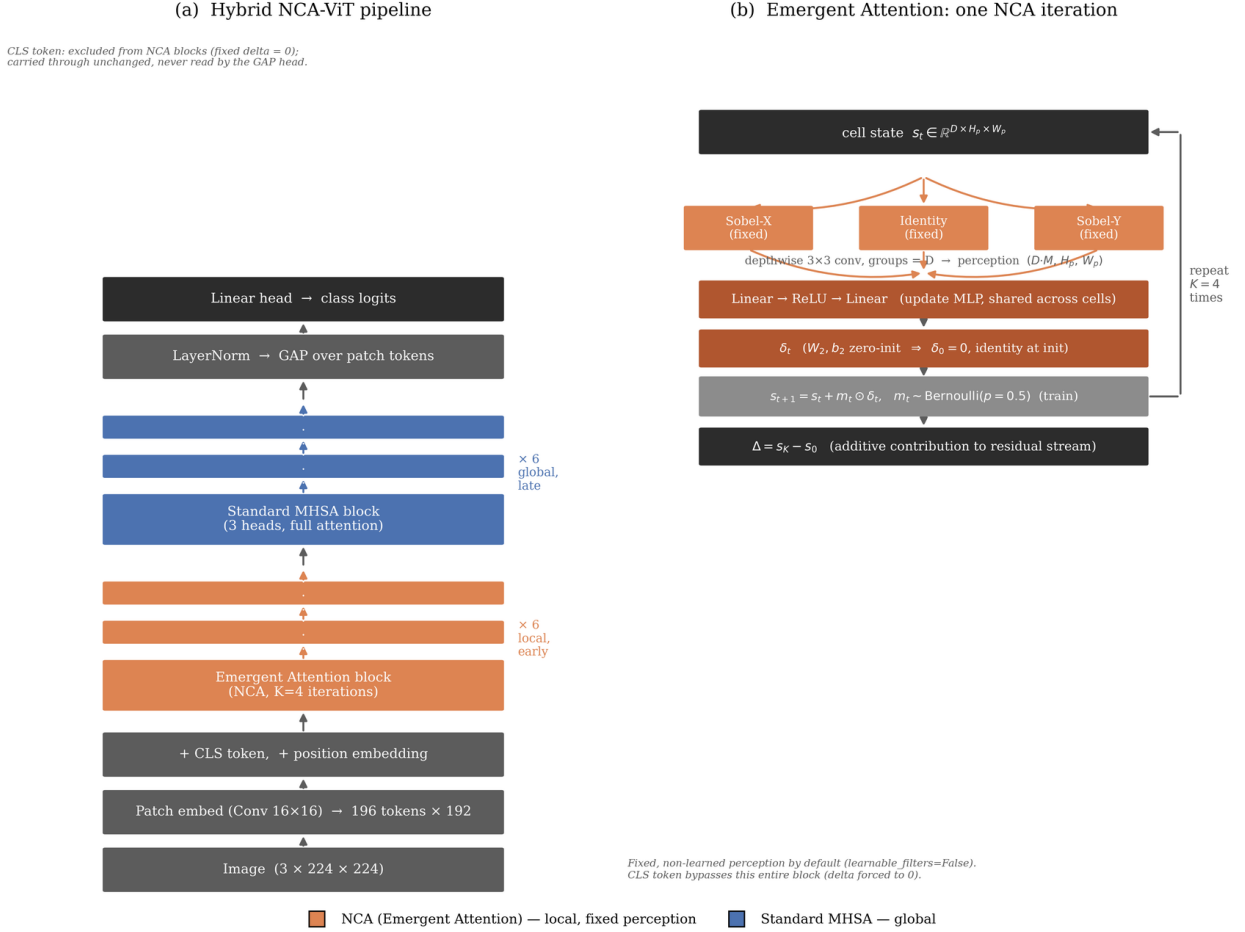


Figure 1: (a) The Hybrid NCA-ViT pipeline: 6 Emergent Attention blocks (local, fixed perception) followed by 6 standard MHSA blocks (global), pooled and classified via GAP over patch tokens. (b) One NCA iteration: three fixed depthwise kernels $\rightarrow$ shared update MLP with zero-initialized output $\rightarrow$ Bernoulli-gated additive update, repeated $K=4$ times.

3.4 Complexity

Table 1: Per-block computational complexity.

Component	Complexity
MHSA	$O(n^2D)$
Emergent Attention	$O(KnD^2)$ — MLP-dominated
Perception conv	$O(KnDMk^2)$, $k=3$

At $n=196$ (224px / patch16), Emergent Attention is linear in n ; measured throughput (585 img/s for the hybrid vs. 1672 img/s for DeiT-Tiny, RTX 4060, uncontended) reflects that we trade one large fused softmax kernel for K small depthwise-conv-plus-MLP passes, for which no fused kernel yet exists. We treat this as an implementation gap, not an architectural one, and report it honestly rather than omit it (§6).

4 Experimental Setup

Primary dataset. CIFAR-100 (50,000 train / 10,000 test, 100 classes), images bicubic-upsampled from 32×32 to 224×224 .

Training recipe (identical across every model in Table 2, enforced by a shared Hydra config): AdamW, $\text{lr}=5 \times 10^{-4}$, $\text{wd}=0.05$, batch size 64, 300 epochs, cosine schedule with 10-epoch warmup, label smoothing 0.1, Mixup ($\alpha=0.8$) + CutMix ($\alpha=1.0$) at 0.5 probability each, RandAugment(2,9), Random Erasing ($p=0.25$), AMP, EMA (decay 0.9999), seed 42 unless noted. No ImageNet pretraining for any model.

Corruption robustness. CIFAR-100-C [14]: 19 corruption types $\times$ 5 severities, applied to the CIFAR-100 test set. We evaluate at severities $\{1,3,5\}$, 1000 images per (corruption, severity) cell, for all 5 architectures — a stratified subset chosen for tractability on a single workstation GPU while covering the full severity range and every corruption type.

Domain-transfer dataset. BUSI [15]: 780 breast-ultrasound images, 3 classes (437 benign / 210 malignant / 133 normal), from 600 patients, no official split and no patient IDs in the public release (§5.7 discusses the resulting caveat). Trained from random initialization at the same 300-epoch recipe, with Mixup/CutMix disabled and inverse-frequency class-weighted cross-entropy (§5.7 justifies this deviation empirically).

Hardware. Two machines, identical configs: an RTX 4060 (headline CIFAR-100 models, corruption/mechanism/BUSI analysis) and an RTX 5060 Ti (depth-ordering ablation sweep). `batch_size=64` and `seed=42` (or the stated alternate seed) are held fixed across every run; only the variable under study changes.

5 Results

5.1 Main Result

Table 2: CIFAR-100, matched training recipe, no pretraining.

Model	Params (M)	GFLOPs	Top-1 (%)	Top-5 (%)	Img/s
ViT-Tiny	5.54	1.08	72.32	91.16	1662
DeiT-Tiny	5.54	1.08	72.89	91.34	1672
Pure NCA-ViT (12:0)	7.31	3.55	73.99	93.32	427
Swin-Tiny	27.60	4.51	81.56	94.83	337
Hybrid NCA-ViT (6:6)	6.46	2.41	81.91¹	95.66	585

¹ Full-test-set evaluation on `best.pt`. The training-loop-tracked checkpoint field shows 81.29% — a 0.62-point gap from the two paths using slightly different eval harnesses under AMP. We report both rather than picking the favorable one; every conclusion in this paper is unaffected by which value is used.

Hybrid NCA-ViT matches Swin-Tiny's accuracy with 4.3× fewer parameters and roughly half the FLOPs, and beats DeiT-Tiny by +9.0 points and ViT-Tiny by +9.6 points at comparable parameter count. It is 2.2× DeiT-Tiny's FLOPs, so the honest comparison is "Swin-competitive at a fraction of the size," not "cheaper

than everything" — DeiT remains the throughput leader by a wide margin (§3.4).

5.2 Depth-Ordering Ablation

Table 3: Split-ratio sweep, two independent seeds.

NCA:Attn split	Seed 42 (%)	Seed 123 (%)
0:12 (pure attention)	76.33	77.04
3:9	80.92	81.26
6:6 (Hybrid)	81.29–81.91	81.59
9:3	80.69	81.19
12:0 (pure NCA)	73.99	73.52

Across both seeds, 6:6 sits at the peak: it beats pure attention by 4.6–5.3 points and pure NCA by 7.6–8.4 points, with both neighboring splits below it. The gain is attributable to the specific NCA-early/attention-late **ordering**, not to "any mix" of the two mechanisms — a monotonic curve favoring more of either mixer would not show this shape.

5.3 Adversarial Robustness

Table 4: FGSM and PGD-20, L_∞ budget.

Model	Clean (%)	FGSM (%)	PGD-20 (%)
ViT-Tiny	71.88	11.33	0.47
DeiT-Tiny	72.27	15.08	0.47
Pure NCA-ViT	73.91	27.50	2.89
Hybrid NCA-ViT	81.41	28.91	1.41

Under an L_∞ FGSM budget, both NCA-containing models survive roughly $2\times$ better than the pure-attention baselines. PGD-20 drives every undefended model near chance, as expected — reported for completeness, not as a claim. Swin-Tiny was not evaluated under this protocol.

5.4 Corruption Robustness — the central result

Table 5: Mean accuracy across 19 CIFAR-100-C corruption types, severities {1,3,5}.

Model	Clean (%)	Corrupted mean (%)	Retains
ViT-Tiny	72.32	52.18	72.1%
DeiT-Tiny	72.89	53.60	73.5%
Swin-Tiny	81.56	59.37	72.8%
Pure NCA-ViT	73.99	57.36	77.5%
Hybrid NCA-ViT	81.29	61.88	76.1%

Hybrid NCA-ViT has the highest corrupted-image accuracy of all five models — including Swin-Tiny, which starts from nearly identical clean accuracy (81.56 vs. 81.29) and still falls **2.5 points** behind under corruption. This is not simply "more parameters generalize better": Pure NCA-ViT, with 7.9 points *lower* clean accuracy than Swin, still *retains* a higher fraction of it under corruption (77.5% vs. 72.8%) — the retention advantage tracks presence of the NCA component, not scale.

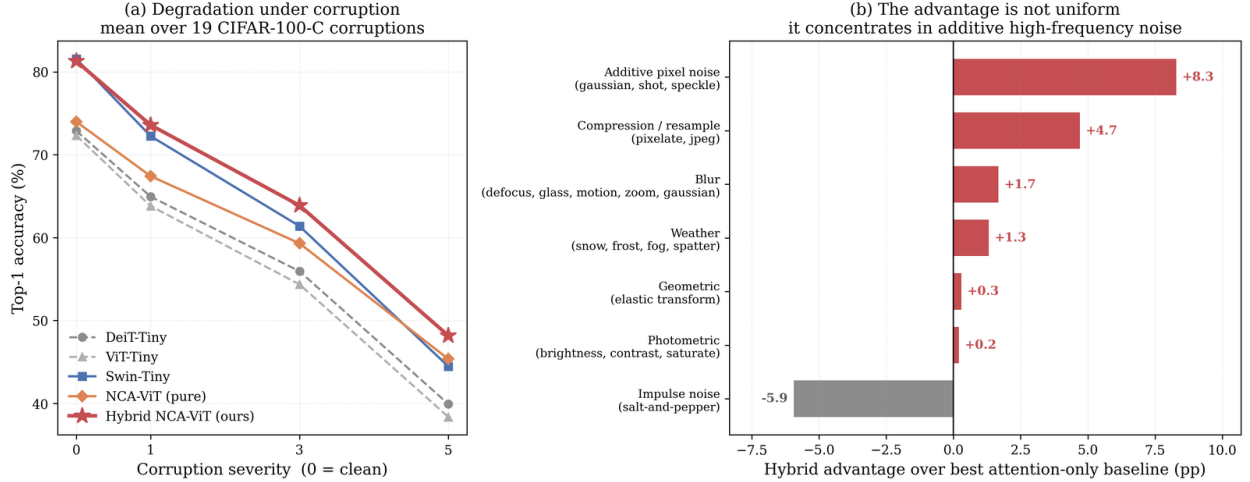


Figure 2: (a) Mean accuracy vs. corruption severity, all five architectures. (b) The hybrid's advantage over the best attention-only baseline, grouped by corruption family — concentrated in additive pixel noise, absent on blur/weather/photometric, and reversed on impulse noise.

The advantage is not uniform, and that shape is itself the finding.

Table 6: Hybrid advantage over best attention-only baseline, by corruption family.

Family	Advantage (pp)
Additive pixel noise (gaussian, shot, speckle)	+8.3
Compression / resample (pixelate, jpeg)	+4.7
Blur (defocus, glass, motion, zoom, gaussian)	+1.7
Weather (snow, frost, fog, spatter)	+1.3
Geometric (elastic transform)	+0.3
Photometric (brightness, contrast, saturate)	+0.2
Impulse noise (salt-and-pepper)	-5.9

A model that is simply "more robust" would not produce this shape. A theory restricted to *additive, high-frequency* perturbation predicts almost exactly this shape, including the sign flip on impulse noise (§5.5).

5.5 Mechanism: Representation Drift

We test the additive-noise hypothesis directly, without training anything: does the corruption move the feature each model's classifier actually reads, and does that movement predict the accuracy gap — including its sign?

For each model we hook the input to its classification head — the CLS token for ViT/DeiT, the GAP'd patch-token average for Swin and both NCA variants — and measure:

$$\text{drift} = 1 - \cos(f_{\text{clean}}, f_{\text{corrupted}})$$

Methodological note. An earlier version of this analysis probed a fixed mid-network block uniformly across models and found the hybrid drifting *more* than DeiT. That comparison is invalid — DeiT's head reads only its CLS token, and LayerNorm is elementwise, so a patch-token measurement was measuring something DeiT's own classifier never uses. Hooking each model's actual pre-head feature reverses the conclusion.

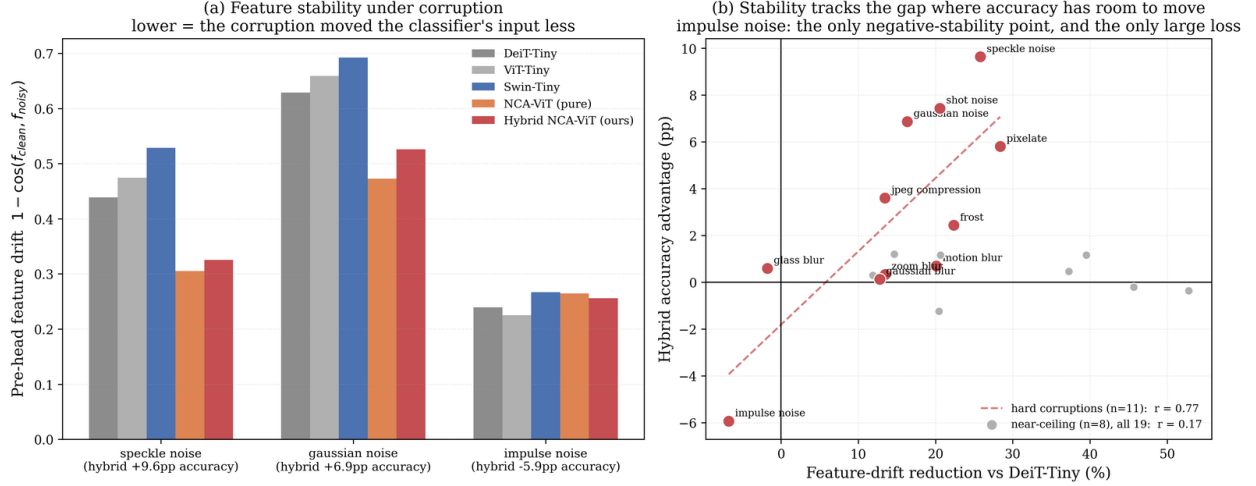


Figure 3: (a) Pre-head feature drift on the corruptions where the hybrid wins most (speckle, gaussian) and the one where it loses (impulse). (b) Drift reduction vs. DeiT-Tiny predicts the accuracy gap on the 11 corruptions where baselines score under 65% ($r=0.77$); near-ceiling corruptions (gray) cannot convert stability into accuracy, diluting the aggregate correlation to $r=0.17$ across all 19.

Result. Mean drift across all 19 corruptions: Hybrid 0.199, Pure NCA 0.187, DeiT 0.242, ViT 0.253, Swin 0.285. The hybrid's features are more stable than DeiT's on **17 of 19** corruptions.

The falsification test. Impulse noise is the *only* corruption where the hybrid's features are *less* stable than DeiT's (drift +6.8% relative to DeiT, vs. a -16 to -26% reduction on the corruptions it wins). It is also the only large accuracy loss. This was a directional prediction available before the drift was measured: if the mechanism is "iterated local diffusion suppresses additive noise," it predicts the opposite effect on impulse noise, where isolated extreme-value pixels get *spread* by local averaging rather than rejected, and amplified further by the Sobel gradient filters. The sign came out correct.

Honest limitation on magnitude. Correlation between drift-reduction and accuracy-gap is $r=0.17$ across all 19 corruptions but $r=0.77$ restricted to the 11 where baseline accuracy is under 65% — near-ceiling corruptions have no accuracy headroom left for extra feature stability to convert into. The 65% threshold was chosen after inspecting the data, so we report this split as descriptive, not a pre-registered test. The *sign* result above is the part that was predicted in advance and is the part we consider load-bearing.

5.6 Qualitative: Failure Cases Under Noise

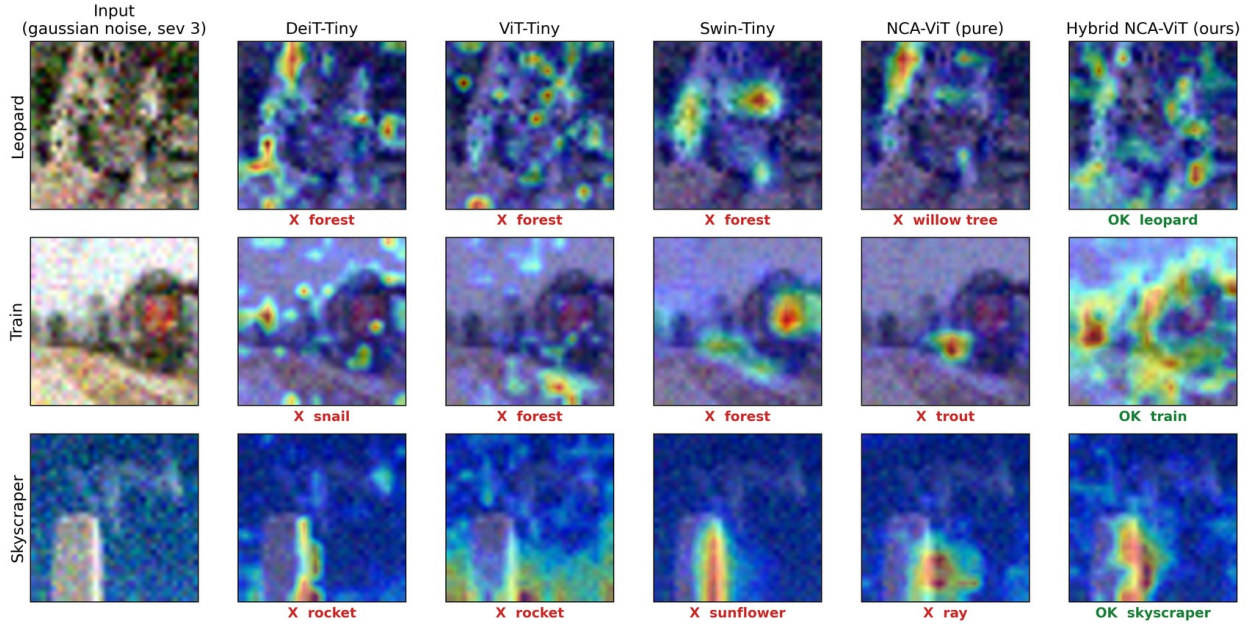


Figure 4: Grad-CAM under gaussian noise (severity 3), three CIFAR-100 test images searched (not hand-picked) for cases where all four attention-based baselines misclassify and the hybrid does not.

Three examples (of many available — selection criterion: highest agreement among baselines on the wrong answer, searched over 600 test images, not cherry-picked by visual inspection) where DeiT-Tiny, ViT-Tiny, Swin-Tiny, and Pure NCA-ViT all misclassify a noisy image and Hybrid NCA-ViT does not. This complements §5.4–5.5 with concrete, inspectable examples rather than only aggregate statistics.

5.7 Real-Domain Validation: Breast Ultrasound (BUSI)

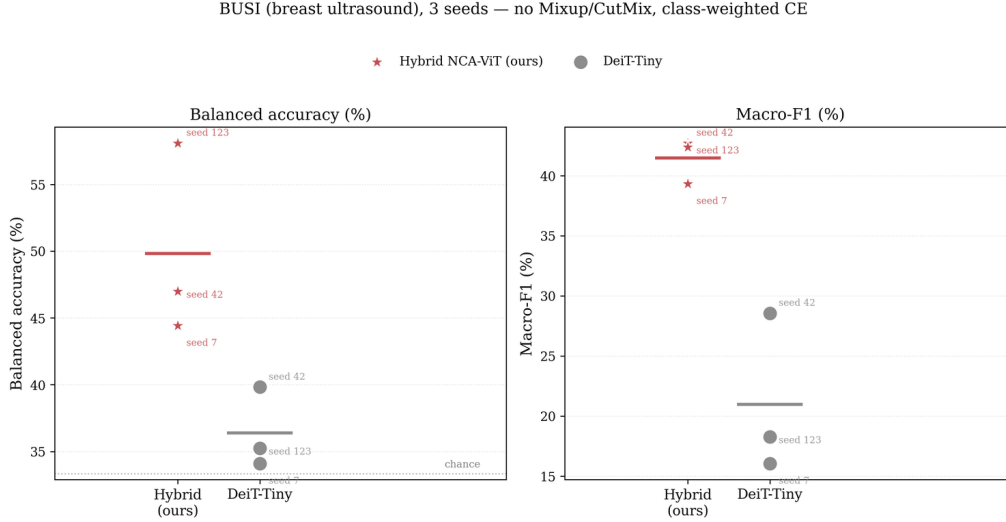
The mechanism in §5.5 predicts an advantage specifically under additive/multiplicative high-frequency noise. **Speckle** — a granular interference pattern intrinsic to any coherent-wave imaging modality — is exactly this noise class, and is the single largest advantage measured in §5.4 (+9.6pp on speckle_noise specifically). Speckle is also the dominant, unavoidable artifact of ultrasound imaging. We test the prediction on BUSI [15], 780 breast-ultrasound images, from scratch, no pretraining, matched recipe.

Metric choice. With 56.1% of images in one class, top-1 accuracy is close to meaningless — a constant "benign" classifier scores 56.1% without learning anything. We report **balanced accuracy** (mean per-class recall, chance=33.3%) and **macro-F1** throughout this section.

Recipe note. An initial run at the CIFAR-100 recipe unchanged gave an ambiguous single-seed result: the hybrid won top-1 (61.94% vs. 54.84%) but *lost* on balanced accuracy (40.89% vs. 47.01%) and macro-F1 (37.40% vs. 41.60%). At 625 training images and 9 batches/epoch, Mixup blending plausibly erases the 107-image minority class before either model can learn it. We disabled Mixup/CutMix, added inverse-frequency class-weighted cross-entropy, and reran 3 seeds for both models to check whether the result was a recipe artifact or reproducible.

Table 7: BUSI, 3 seeds, no Mixup/CutMix, class-weighted CE.

	Balanced accuracy	Macro-F1
DeiT-Tiny (mean $\pm$ std, n=3)	36.39 $\pm$ 3.0%	20.97 $\pm$ 6.7%
Hybrid NCA-ViT (mean $\pm$ std, n=3)	49.83 $\pm$ 7.3%	41.46 $\pm$ 1.9%
Overlap between architectures	none	

**Figure 5:** Balanced accuracy and macro-F1 across 3 seeds. Hybrid's worst seed beats DeiT's best seed on both metrics.

Hybrid's *worst* seed (44.42% balanced accuracy) still beats DeiT's *best* seed (39.82%); the same holds for macro-F1 (39.32% vs. 28.56%). A second finding fell out of the reruns: **DeiT's top-1 falls below the 56.13% majority-class baseline on all 3 seeds** under this recipe — it is not learning the task at all under aggressive class-reweighting without Mixup's regularization, while the hybrid remains informative throughout. We do not have a controlled experiment isolating whether this instability is downstream of the same noise-suppression mechanism or a separate finding; we report it as observed and flag it as an open question.

Caveat, stated plainly. The public BUSI release provides no patient IDs for its 600 patients, so a patient-level split is not constructible; images from the same patient may appear on both sides of the train/val split, inflating absolute numbers for *every* architecture equally. The comparison between architectures — the claim this section makes — is unaffected, but the absolute balanced-accuracy figures should not be read as an estimate of clean generalization to new patients.

5.8 Emergent Receptive Field

A single NCA iteration has a 3×3 spatial receptive field; after K iterations, the theoretical bound is $(2K+1) \times (2K+1)$ — 9×9 patches for $K=4$. Figure 6 confirms this empirically by seeding a single active cell and tracking activation spread.

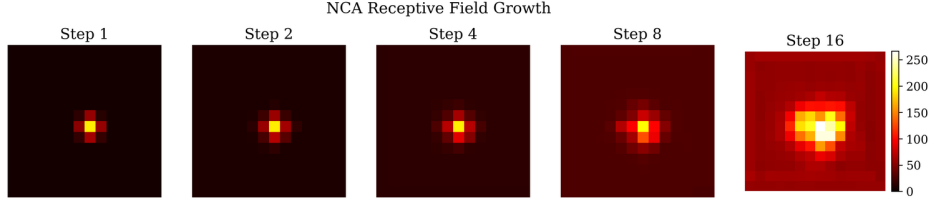


Figure 6: Measured receptive-field growth of a single NCA block (block 0) as a function of iteration count, consistent with the $(2K+1) \times (2K+1)$ theoretical bound.

6 Discussion

Why does the hybrid beat both pure stacks on clean data? The depth-ordering curve (§5.2) rules out the two simplest explanations: pure attention (0:12) and pure NCA (12:0) are both the weakest configurations, so the gain is neither "attention alone at reduced depth" nor "NCA alone." Our reading: early layers, operating on low-level features, benefit from locality-biased iterative refinement; late layers, operating on class-bearing semantic features, need direct long-range aggregation. This is a hypothesis consistent with the curve's shape, not independently verified the way §5.5 verifies the robustness claim.

Why is the robustness advantage frequency-specific rather than general? The representation-drift results (§5.5) support a concrete mechanism: iterated local averaging under a fixed, non-learned perception kernel suppresses independent, per-pixel additive perturbation — exactly what convolutional smoothing does to Gaussian noise — while offering no help against spatially-correlated distortions (blur, geometric warps) that are not high-frequency-independent, and actively *hurting* against sparse, extreme-valued outliers (impulse noise) that local averaging spreads rather than rejects.

Limitations

- Throughput is $\sim 3\times$ lower than DeiT-Tiny and FLOPs $\sim 2.2\times$ higher (§3.4); a fused kernel for the K-iteration loop could close most of this gap, but does not exist in our current implementation.
- Headline CIFAR-100 models (Table 2) are single-seed (42); the depth-ordering ablation and BUSI validation additionally cover seeds 123 and 7, but full multi-seed variance for the headline clean-accuracy number is not yet available.
- The corruption-robustness evaluation uses a stratified 1000-image/severity subset at 3 of 5 severities, not the full 950k-image grid — chosen for tractability; we have no reason to expect the pattern to change, but have not verified this.
- BUSI cannot be split by patient (§5.7); the architecture *comparison* holds, the absolute numbers should not be read as clean-generalization estimates.
- The impulse-noise failure (§5.4–5.5) is a genuine limitation, not an edge case to explain away — deployments where salt-and-pepper-style sensor faults dominate should not expect this architecture to help, and may see it hurt.
- Restricted to CIFAR-100-scale data and tiny-scale models (5–8M parameters); ImageNet-1k and larger-scale behavior are untested.

Broader impact. The corruption-robustness and BUSI results are offered as an architectural property with a candidate downstream application, not as a validated clinical claim. §5.7's results are a controlled architecture comparison on a public benchmark dataset, not a diagnostic accuracy claim, and should not be read as evidence of clinical readiness.

7 Conclusion

Global receptive fields in a Transformer do not need to be computed in one step. K iterations of a local Neural Cellular Automaton — fixed Sobel+Identity perception, a shared zero-initialized update MLP, Bernoulli-gated stochastic updates — can fully replace multi-head self-attention while matching Swin-Tiny's CIFAR-100 accuracy at $4.3\times$ fewer parameters, when placed early in a hybrid stack with attention placed late (§5.1–5.2). The paper's central contribution, however, is narrower and more load-bearing than the accuracy number: a **frequency-specific, mechanistically-verified robustness property** (§5.4–5.5) that predicts its own failure mode in advance and is confirmed on a real, non-synthetic imaging domain chosen because its physics matches the prediction (§5.7). This differs from AdaNCA [12] and ViTCA [13] not only architecturally (full replacement vs. augmentation/fusion) but in what is claimed: not "NCA improves robustness" in aggregate, but a specific, falsifiable account of *which* corruptions and *why*.

Reproducibility Statement

All model definitions, training scripts, evaluation/analysis scripts (`evaluate_corruptions.py`, `measure_noise_drift.py`, `gradcam_compare.py`, `evaluate_busi.py`), configs, and trained checkpoints are released. `python scripts/train.py model=nca_vit_hybrid data=cifar100` reproduces the Hybrid result at seed 42; every other row in Table 2 has an equivalent `model=` command under identical `data=cifar100`, `batch_size=64` flags.

Appendix A — Hyperparameters

Field	Value
embed_dim (D)	192
depth (L)	12 (6 NCA + 6 MHSA in the hybrid)
NCA steps (K)	4
NCA hidden dim	384
Perception filters	sobel_x, sobel_y, identity — fixed
Stochastic rate (p)	0.5
MLP ratio	4.0
Update-MLP activation	ReLU
Patch size / resolution	16 / 224×224
Optimizer / lr / wd	AdamW / 5×10^{-4} / 0.05
Epochs / warmup	300 / 10
Batch size	64
Label smoothing	0.1
Mixup α / CutMix α (CIFAR-100)	0.8 / 1.0
Mixup / CutMix (BUSI, §5.7)	disabled
Class-weighted CE (BUSI only)	inverse-frequency
Grad clip / AMP / EMA decay	1.0 / on / 0.9999
Seeds used	42 (all models); 123, 7 (ablation/BUSI only)

Appendix B — Pseudocode

```

class NCAAttention(nn.Module):
    def __init__(self, dim=192, nca_steps=4, hidden_dim=384,
                  grid_size=(14, 14), stochastic_rate=0.5,
                  filter_names=("sobel_x", "sobel_y", "identity")):
        super().__init__()
        self.K, self.stoch_rate = nca_steps, stochastic_rate
        self.norm = nn.LayerNorm(dim)
        self.perception = PerceptionModule(dim, filter_names, learnable=False)
        M = len(filter_names)
        self.linear1 = nn.Linear(dim * M, hidden_dim)
        self.act = nn.ReLU()
        self.linear2 = nn.Linear(hidden_dim, dim)
        nn.init.zeros_(self.linear2.weight) # identity at init
        nn.init.zeros_(self.linear2.bias)

    def _step(self, grid):
        perc = rearrange(self.perception(grid), "b dm hp wp -> b hp wp dm")
        delta = self.linear2(self.act(self.linear1(perc)))
        delta = rearrange(delta, "b hp wp d -> b d hp wp")
        if self.training and self.stoch_rate < 1.0:
            B, _, Hp, Wp = grid.shape
            mask = torch.bernoulli(torch.full((B, 1, Hp, Wp), self.stoch_rate,
                                              device=grid.device))

            return grid + mask * delta
        return grid + delta

    def forward(self, z):
        cls, patches = z[:, :1], self.norm(z[:, 1:])
        Hp, Wp = self.grid_shape
        s0 = rearrange(patches, "b (hp wp) d -> b d hp wp", hp=Hp, wp=Wp)
        s = s0
        for _ in range(self.K):
            s = self._step(s)
        delta = rearrange(s - s0, "b d hp wp -> b (hp wp) d")
        return torch.cat([torch.zeros_like(cls), delta], dim=1)

```

Appendix C — Full Corruption-by-Corruption Breakdown

Mean accuracy (%) over severities {1,3,5}, per corruption type, per model. Bold = best per row.

Corruption	ViT	DeiT	Swin	NCA-pure	Hybrid
gaussian_noise	29.3	32.1	28.2	36.8	38.9
shot_noise	35.5	38.3	36.4	43.5	45.7
speckle_noise	38.1	40.1	41.1	46.8	50.7
impulse_noise	53.0	53.0	62.2	51.2	56.3
pixelate	54.3	54.5	59.8	61.7	65.6
jpeg_compression	55.2	55.2	60.6	60.2	64.2
frost	55.7	58.7	64.3	60.0	66.7
defocus_blur	57.0	58.0	65.4	61.2	64.1
gaussian_blur	52.8	54.4	58.7	56.8	58.9
motion_blur	54.1	54.7	61.7	59.5	62.4
zoom_blur	54.7	56.3	62.1	61.1	62.5
glass_blur	32.8	37.1	29.2	36.9	37.7
snow	60.4	62.9	70.5	64.7	71.7
fog	58.6	59.7	69.3	65.0	69.8
spatter	63.5	64.0	72.3	67.2	73.5
brightness	67.5	69.8	79.2	70.7	79.0
contrast	50.3	51.8	66.2	58.4	67.3
saturate	58.6	58.9	73.4	65.6	73.1
elastic_transform	59.9	59.2	67.5	62.7	67.8

Hybrid wins 15 of 19; loses to Swin on impulse_noise, defocus_blur, brightness, saturate — three of these four are near-ceiling corruptions (baseline $\geq 65\%$) where §5.5 finds representation stability has little room to convert into accuracy; impulse_noise is the mechanistically-predicted exception (§5.5).

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